INCH-POUND

MIL-DTL-32729
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DETAIL SPECIFICATION

CONVERTER, AIRCRAFT, UNIVERSAL SERIAL BUS CHARGING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general requirements for converters that convert electrical power in aircraft from nominal 115 volts root mean square (VRMS) alternating current (AC), 400 Hertz (Hz), AC single-phase (see 6.7.2 and 6.7.3), or 28 volts direct current (VDC) (see 6.7.5) to nominal 5 volts VDC. The converter's intended use is to provide battery charging power to devices using Universal Serial Bus (USB) 5 VDC for battery charging.

1.2 Classification. Converters are of the following types:

Type AC. Type AC converters convert 115 VRMS AC 400 Hz input to 5 VDC output.

Type DC. Type DC converters convert 28 VDC to 5 volts VDC output.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specification, standards, and handbooks. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to LKE-NAVAIRDocComments@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-202-111	-	Flammability (External Flame)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-464	-	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-704	-	Aircraft Electric Power Characteristics
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Galvanic Compatibility of Electrically Conductive Materials

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-704-1	-	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics (Part 1 of 8 Parts)
MIL-HDBK-704-2	-	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics Single Phase, 400 HZ, 115 Volt (Part 2 of 8 Parts)
MIL-HDBK-704-3	-	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics Three Phase, 400 HZ, 115 Volt (Part 3 of 8 Parts)
MIL-HDBK-704-4	-	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics Single Phase, Variable Frequency, 115 Volt (Part 4 of 8 Parts)
MIL-HDBK-704-5	-	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics Three Phase, Variable Frequency, 115 Volt (Part 5 of 8 Parts)
MIL-HDBK-704-6	-	Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics Single Phase, 60 HZ, 115 Volt (Part 6 of 8 Parts)
MIL-HDBK-704-7	-	Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics 270 VDC (Part 7 of 8 Parts)

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MIL-HDBK-704-8 - Guidance for Test Procedures for Demonstration of Utilization Equipment Compliance to Aircraft Electrical Power Characteristics 28 VDC (Part 8 of 8 Parts)

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM90415 - Nut, Self-Locking, Steel, 160 KSI, 450 Degrees F, 12 Point, Captive Washer

(Copies of this document are available from https://www.aia-aerospace.org.)

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

ECIA EIA-364-26 - Salt Spray Test Procedure for Electrical Connectors, Contacts and Sockets

(Copies of this document are available online at https://www.ecianow.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 15434 Barcode Specification

(Copies of this document are available online at https://webstore.ansi.org.)

JOINT ELECTRON DEVICE ENGINEERING COUNCIL (JEDEC)

JESD471 - Symbol and Label for Electrostatic Sensitive Devices

(Copies of this document are available from https://www.jedec.org.)

RTCA, INCORPORATED

RTCA/DO-160 - Environmental Conditions and Test Procedures for Airborne Equipment

(Copies of this document are available from https://www.rtca.org.)

SAE INTERNATIONAL

SAE AMS-STD-595/37038 - Colors used in Government Procurement
SAE AMS-M-3171 - Magnesium Alloy, Processes for Pretreatment And Prevention of Corrosion on
SAE AS8879 - Screw Threads - UNJ Profile, Inch Controlled Radius Root with Increased Minor Diameter
SAE AS50881 - Wiring, Aerospace Vehicle

(Copies of these documents are available from https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related

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specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet (see 6.10). In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Converters furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable Qualified Products List (QPL) before contract award (see 4.2 and 6.3).

3.3 Materials.

3.3.1 Selection of materials, parts, and processes. The materials used shall enable the converter to meet all of the operational and environmental performance requirements of this specification and the applicable specification sheet (see 6.4.2 and 6.4.3).

3.3.2 Moisture resistance. Non-metal materials, including plastics, ceramics, fabrics, and protective finishes, shall be moisture resistant. Non-metal materials and parts may be treated to conform to this requirement. Electrical connector selection shall be environment resistant for a 500 hour salt spray environment (see 3.4.2.1.1.1).

3.3.3 Corrosion resistance. All metals used in the converter construction shall be processed and protected to resist corrosion (see 6.4.2.3). The use of magnesium is prohibited, unless specifically approved for each application by the procuring activity. Where approved, magnesium alloy parts shall be surface treated for protection against corrosion in accordance with SAE AMS-M-3171. Dissimilar metals, when used in contact with each other, shall be protected against electrolytic corrosion in accordance with MIL-STD-889. Corrosion protection applied to dissimilar materials, such as plating and coatings, shall withstand the environmental tests herein.

3.3.4 Contamination by fluids (see 4.6.3.8). The converter shall show no deterioration at normal operating temperatures due to wetting from aviation fuel (JP-4/JP-5), fire extinguishing compounds, de-icing fluids, aircraft washing fluids, hydraulic fluids, and lubricating oils (see 6.8).

3.3.5 Toxicity and fire resistance (see 4.6.3.9). Materials used shall be flame-resistant, shall not support combustion, and shall be non-toxic when exposed to flame as well as when used under all operating and environmental conditions herein.

3.3.6 Recycled, recovered, or environmentally preferable materials. The use of toxic chemicals, hazardous substances, and ozone-depleting chemicals shall be avoided in the preparation and production of equipment (see 6.4.2.1). Recycled, recovered, or environmentally preferable materials shall be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.3.7 Color. The converter shall be finished in accordance with SAE AMS-STD-595/37038 black or as identified in the specification sheet.

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3.4 Design and construction.

3.4.1 General mechanical and electrical design. The converter shall be constructed with parts and materials to provide the specified performance, reliability, and service life under all environmental and operating conditions specified herein. Electronics parts shall be selected to enable the converter to provide the specified performance, reliability, and service life (see 6.4.3). Static discharge control shall be provided for the protection of electronic devices during assembly and handling (see 6.4.3.4). Dielectrics used for electrical isolation shall be adequate to prevent breakdown under all specified environmental conditions for the life of the equipment. Equipment shall be free from irregularities or defects that could degrade performance or durability. Standard design practices shall be followed (see 6.4).

3.4.2 Electrical design.

3.4.2.1 Electric connections.

3.4.2.1.1 External electric connections. The equipment shall be designed so that external connections and wiring can be made in accordance with the wiring installation, ground return, and connector requirements of SAE AS50881. The equipment structure shall not be used as a current path except for electromagnetic (radio noise) shielding. External connections necessary for appropriate input and output shall be as shown on the applicable specification sheet.

3.4.2.1.1.1 Connectors. The input and output external connectors shall be selected so that the contacts on the "live" or "hot" side of the connection are socket type rather than pin type to minimize personnel hazard and prevent accidental shorting of live circuits when the connector is unmated. Only connectors that withstand a minimum of 500 hours salt spray, when tested per ECIA EIA-364-26 or equivalent shall be used.

3.4.2.1.2 Internal electric connections. Internal wiring and connections shall enable the converter to meet all of the operational and environmental performance requirements of this specification and the applicable specification sheet (see 6.4.3.2).

3.4.2.2 USB charging electrical power routing. The converter external connectors, electrical connections, chassis ground and current returns shall be specified in the applicable specification sheet. See 6.1.1 for routing guidance.

3.4.2.3 Parts de-rating. Electronic parts, including semiconductors, shall be de-rated in accordance with Table X. Steady state operating parameters of parts shall not exceed the derated values specified unless approved by the qualifying activity. Requests for qualifying activity approval for de-rating exceedances shall be accompanied by an analysis of the reliability, weight, cost and performance impact. Operating parameters of parts shall not exceed limits specified by the manufacturer. See 6.4.3.

3.4.3 Mechanical design.

3.4.3.1 Moving parts. The converter shall not have any moving parts.

3.4.3.2 Shock mounts. Shock mounts or vibration isolators shall not be used.

3.4.3.3 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation and performance.

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3.4.3.4 Mechanical connections, fasteners, inserts, and hardware retention. Screw threads shall be unified inch standard series UN, UNR, and UNJ (see 6.4.2.2). UNJ screw threads shall conform to the requirements of SAE AS8879. All internal and external threaded parts shall be positively locked (self-locking nuts, safety wiring, or acceptable alternative methods used to prevent loosening of threaded parts). Staking shall not be used. Internal devices with mechanical adjustment provisions shall be locked and sealed during production. Stud nuts shall be self-locking steel nuts with a captive washer in accordance with NASM90415.

3.4.3.5 Cooling. The converter shall be self-cooled. No conduction cooling to the converter environment shall be allowed. Safe-operating equilibrium temperature shall be maintained for the converter over the specified range of operating conditions, and the rated temperature of individual converter components shall not be exceeded.

3.4.3.6 Operating position. The converter shall operate in any position, unless otherwise specified in the applicable specification sheet.

3.4.3.7 Size and weight. The dimensions and weight shall be as required by the applicable specification sheet. The weight and dimensions shown on the applicable specification sheet shall include the weights and dimensions of all auxiliary apparatus necessary to make the converter conform to this specification and the applicable specification sheet. Where only maximum and minimum dimensions are shown, the part, including all protrusions, shall be contained within the outline shown.

3.5 Performance characteristics. The converter shall deliver power in accordance with the requirements specified herein and the applicable specification sheet.

3.5.1 Electrical (see 4.6.2). The converter shall utilize electric power in accordance with MIL-STD-704 (see 6.1.4) and shall be in accordance with the utilization equipment requirements of MIL-STD-704. The converter shall only provide electrical power as required by this specification and applicable specification sheet and shall not be used for digital data transfer. The use of high-speed data cabling and connectors is not required.

3.5.1.1 Type AC converter input power. The converter shall meet the requirements of this specification and the applicable specification sheet when supplied with power as specified herein, as it applies to an aircraft MIL-STD-704 compliant electrical system. The nominal input voltage is 115 VRMS (volts root mean square) alternating voltage (see 6.7.2 and 6.7.3) at 400 Hz input.

3.5.1.1.1 Steady-state input voltage.

3.5.1.1.1.1 Full performance. The converter shall meet full performance output requirements (see 3.5.1.3.1.1) with any combination of the following input characteristics:

a. Steady-state input voltage between 108 VRMS and 118 VRMS, unless otherwise specified by the applicable specification sheet.

b. Steady-state input frequency between 380 Hz and 420 Hz (see 6.1) , unless otherwise specified by the applicable specification sheet.

c. Input waveform (see 4.6.2.8.2) total distortion of 8 percent of the fundamental or less, with individual harmonic frequency components below 5 percent individual.

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d. Input voltage amplitude modulation of 2.5 VRMS.

3.5.1.1.1.2 Abnormal and emergency operation. The converter shall not be damaged during aircraft electrical system abnormal and emergency operation (see 6.7.1, 6.7.6, and MIL-STD-704) and shall return to full performance (see 3.5.1.1.1.1) after aircraft electrical system normal operation (see 6.7.8, and MIL-STD-704). The converter's return to normal operation shall be verified when each of the following characteristics is separately substituted for the corresponding characteristic listed for full performance listed in 3.5.1.1.1.1:

a. Steady-state input voltage between 100 VRMS and 108 VRMS, unless otherwise specified by the applicable specification sheet.

b. Steady state input voltage between 118 VRMS and 125 VRMS, unless otherwise specified by the applicable specification sheet.

c. Non-harmonic frequency components (including modulated sidebands) shall meet Figure 1, Curve B limits unless otherwise specified by the applicable specification sheet.

3.5.1.1.2 Transient input voltage (see 4.6.2.8).

3.5.1.1.2.1 Transient surge susceptibility.

3.5.1.1.2.1.1 Full performance. The converter output shall remain within the limits of 4.75 VDC and 5.25 VDC, when subjected to transient input voltages within the limits of figure 2 (see 4.6.2.8.3.1.1).

3.5.1.1.2.1.2 Abnormal and emergency operation. The converter shall not be damaged by power interrupts or abnormal input surge voltages as specified by the limits of table I (see 4.6.2.8.3.1.2).

3.5.1.1.3 Transient input frequency. The converter shall withstand without damage and shall recover full performance after application of the following frequency transients (see 4.6.2.8.4):

480 Hz for 7 seconds

320 Hz for 7 seconds

The converter output voltage shall remain between 4.75 VDC and 5.25 VDC during the application of the following frequency transients.

450 Hz for 2 seconds

440 Hz for 15 seconds

360 Hz for 15 seconds

350 Hz for 2 seconds

3.5.1.1.4 Input power factor (see 4.6.2.4). The converter input power factor shall be no less than 0.9 lagging when operated from 50 percent to 100 percent rated load.

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3.5.1.2 Type DC converter input power. The converter shall meet the requirements of this specification when supplied with power as specified herein from an aircraft electrical system. The nominal input voltage is 28 VDC.

3.5.1.2.1 Full performance (see 4.6.2.8.3.1.1). The converter shall meet full performance output requirements (see 3.5.1.3.1.1) with any combination of the following input characteristics:

a. Steady state input voltage between 22 VDC and 29 VDC, unless otherwise specified by the applicable specification sheet.

b. Input distortion factor of 0.035 or less.

c. Input voltage ripple of 1.5 volts peak to mean or less.

3.5.1.2.2 Abnormal and emergency operation (see 4.6.2.8.3.1.2). The converter shall not be damaged during aircraft electrical system abnormal and emergency operation (see 6.7.1, 6.7.6, and MIL-STD-704) and shall return to full performance (see 3.5.1.1.1.1) after aircraft electrical system normal operation (see 6.7.8) and MIL-STD-704). The converter's return to normal operation shall be verified when each of the following characteristics is separately substituted for the corresponding characteristic listed for full performance listed in 3.5.1.2.1:

a. Steady state input voltage between 20 VDC and 22 VDC, unless otherwise specified by the applicable specification sheet.

b. Steady state input voltage between 29 VDC and 31.5 VDC, unless otherwise specified by the applicable specification sheet.

c. Distortion factor shall meet Figure 3 limits unless otherwise specified by the applicable specification sheet.

3.5.1.2.3 Transient input voltage (see 4.6.2.8).

3.5.1.2.3.1 Transient surge susceptibility.

3.5.1.2.3.1.1 Full performance. The converter output shall remain within the limits of 4.75 VDC and 5.25 VDC, when subjected to transient input voltages within the limits of Figure 4 (see 4.6.2.8.3.1.1).

3.5.1.2.3.1.2 Abnormal and emergency operation. The converter shall not be damaged during aircraft electrical system abnormal and emergency operation (see 6.7.1, 6.7.6, and MIL-STD-704) and shall return to full performance (3.5.1.2.1) after aircraft electrical system normal operation (see 6.7.8, and MIL-STD-704). The converter shall not be damaged by power interrupts or abnormal input surge voltages as specified by the limits of table II (see 4.6.2.8.3.1.2).

3.5.1.3 Output power.

3.5.1.3.1 Voltage regulation. The converter shall have a nominal output voltage of 5 VDC and maintain the following steady-state output voltages:

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3.5.1.3.1.1 Full performance input voltage (see 3.5.1.1.1.1 or 3.5.1.2.1). Steady state output voltage shall remain between 4.75 VDC and 5.25 VDC. The maximum current limit shall be 5 amps per USB output connection.

3.5.1.3.1.2 Abnormal and emergency operation. The converter shall not be required to operate during aircraft electrical system abnormal and emergency operation (see 6.7.1, 6.7.6, and MIL-STD-704) and shall return to full performance (see 3.5.1.2.1) after aircraft electrical system normal operation (see 6.7.8, and MIL-STD-704). The converter output voltage shall not exceed 9 VDC during aircraft electrical system abnormal and emergency operation.

3.5.1.3.2 Overload and short circuit capacity (see 4.6.2.5). Over an input voltage range of Type AC converter input voltage of 108 to 118 VRMS and input frequency range of 380 Hz to 420 Hz or Type DC converter input voltage of 22 to 29 VDC, the converter shall be capable of operating without damage into any passive load impedance, including a short circuit, (< 0.5 Ohms) indefinitely.

3.5.1.3.3 Single point failure and resultant output voltage (see 4.6.2.6). The output voltage shall remain between 0.3 and 9 VDC for any single point failure.

3.5.1.3.4 Maximum allowable voltage overshoot (see 4.6.2.6). The output maximum allowable overshoot shall be < 6.0 V for any step-change in load current or when the output is turned on or off.

3.5.1.3.5 Maximum allowable voltage undershoot (see 4.6.2.6). The output maximum allowable undershoot shall be > 4.25 VDC for changes in load current from low (0-30 milliamps [mA]) to mid (30-100 mA) or from mid to high (0.1-2.1 A) for a duration not to exceed 10 milliseconds.

3.5.1.3.6 Maximum output ripple voltage (see 4.6.2.6). The output maximum allowable ripple voltage from DC to 20 MHz shall be less than 100 millivolts (mV) (peak to peak) for current less than or equal to 2.1 amps.

3.5.1.4 Efficiency (see 4.6.2.3). When supplied from an input power source having full performance characteristics specified in 3.5.1.1.1.1 or 3.5.1.2.1, the converter efficiency shall be 85 percent from 50 to 100 percent electrical load unless otherwise specified in the applicable specification sheet.

3.5.1.5 Electromagnetic compatibility (see 4.6.2.7). The converter shall be in accordance with the general interface requirements of MIL-STD-461 and with the CE102, CS101, CS114, CS115, CS116, RE102, and RS103 emissions and susceptibility requirements of MIL-STD-461 (see 6.11). The converter shall not interfere with the performance of other aircraft equipment as a result of radiated or conducted electromagnetic interference. In addition, the converter performance shall not be degraded as a result of electromagnetic interference from other aircraft equipment.

3.5.2 Environmental. The converter shall conform to the requirements specified herein when subjected to the environmental conditions, as specified by the tests herein.

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3.5.2.1 Temperature-altitude. The converter shall meet all performance requirements specified herein throughout the temperature-altitude range specified on Figure 5.

3.5.2.2 Acceleration. The converter shall withstand the inertial loads induced by aircraft operation (acceleration, deceleration, and maneuvers) and provide specified performance when subjected to the acceleration test of 4.6.3.2. The output voltage characteristics of the converter shall remain within the limits of 3.5.1.3.1.1 while operating during the acceleration test. There shall be no evidence of physical damage, and the voltage regulation after the test shall be within the limits specified in 3.5.1.3.1.1.

3.5.2.3 Shock. The converter shall withstand the effects of mechanical shock from handling, transport, and aircraft operations, as simulated by the tests of 4.6.3.3.

a. Functional (see 4.6.3.3.1). The output voltage shall not exceed the limits of 4.45 VDC and 5.25 VDC while operating during the functional shock test. There shall be no evidence of mechanical damage, loosening of parts or hardware, or physical distortion of the enclosure. At the conclusion of the test, the voltage regulation shall be within the limits specified in 3.5.1.3.1.1.

b. Crash hazard (see 4.6.3.3.2). The converter shall remain in place and not create a safety hazard during the crash hazard shock test.

3.5.2.4 Vibration. The converter shall withstand the vibration environment specified herein (see 4.6.3.4 and 6.5). The maximum output ripple voltage of the converter measured during the test shall not exceed 100 mV (peak to peak) (see 3.5.1.3.6). The output voltage measured during the test shall remain within the limits specified in 3.5.1.3.1.1. There shall be no evidence of mechanical damage, loosening of parts or hardware, or physical distortion of the enclosure. At the conclusion of the test, the voltage regulation shall be within the limits specified in 3.5.1.3.1.1.

3.5.2.5 Fungus (see 4.6.3.5). The converter shall be constructed of non-nutrient materials that do not support the growth of fungus in aircraft operating environments.

3.5.2.6 Salt fog. The converter shall withstand the effects of operation and storage in maritime salt fog environments when subjected to the test of 4.6.3.6. As a result of the salt fog test, there shall be no evidence of corrosion, peeling, pitting, blistering, or cracking of the finish. At the conclusion of the test, the voltage regulation shall be within the limits specified in 3.5.1.3.1.1.

3.5.2.7 Explosive atmosphere. The converter shall not cause ignition of the surrounding explosive mixture when tested as specified in 4.6.3.7. Seals and coatings providing ignition hazard protection shall not be damaged by the tests in 4.6.3.7.

3.5.2.8 Electrostatic discharge (ESD) (see 4.6.3.10). The converter shall be unharmed by electrostatic discharge through contact with personnel up to 15 kV air discharge as required in MIL-STD-464 for Electrical and Electronic Subsystems.

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3.6 Reliability. Converters shall have a minimum observed Mean Time Between Failure (MTBF) of 6,000 hours when operated under conditions specified herein, unless otherwise specified in the specification sheet (see 6.4.4).

3.7 Markings. Identification of electrostatic discharge sensitive components shall be in accordance with JEDEC JESD471. As a minimum, the nameplates shall include the following information: (1) item nomenclature, (2) manufacturer's identification; (3) original part number, current part number, and sequence (serial) number; (4) Commercial and Government Entity (CAGE) code number; (5) input voltage, current and frequency; (6) output voltage and current; and (7) date of manufacture. The nameplate may include information to track changes that do not require a part number change. Markings shall remain legible after exposure to the environmental tests specified herein.

3.7.1 Unique item identifiers. Converters shall be assigned unique item identifiers (UII) of construct #2 as defined in MIL-STD-130. The UII shall be encoded as a message formatted in accordance with ISO/IEC 15434. The unique identifier shall be formed using the manufacturer's CAGE code and serialization within the original part number. In text form, the unique identifier shall contain only printable characters (consist of the letter "D" followed by the manufacturer's CAGE code followed by the original part number followed by the sequence number.

3.7.2 Mechanical nameplate. Unless otherwise specified in the applicable specification sheet, the nameplates shall be attached using mechanical fasteners and not an adhesive. The converter shall be marked with a unique identification (UID) nameplate. The UID labels shall contain Human Readable Information (HRI) and Machine Readable Information (MRI). The MRI shall be contained in one or more 2D data matrices.

3.7.3 Electrical connections. Markings of external connectors and terminals shall be as required by the applicable specification sheet.

3.8 Workmanship. The converter manufactured and assembled shall be free from irregularities that could degrade performance or durability.

3.9 Burn-in (see 4.6.5). Manufacturing screening of each converter shall be provided by the random vibration and temperature cycling of the burn-in test.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection.

4.2.1 Qualification sample submittal. The qualifying activity may acquire converters for qualification, or unsolicited samples may be submitted to the qualifying activity by manufacturer's seeking qualification (see 6.2 and 6.3). Qualification may require up to four converters. The qualification test schedule matrix is shown in table III and may be modified by the qualifying activity.

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4.2.2 Qualification inspections and tests. Qualification inspection shall consist of the examinations and tests required by this specification and the applicable specification sheet. The manufacturer or an independent laboratory may conduct qualification testing with approval from the qualifying activity.

a. Laboratory qualification test - The examinations and tests listed in table III shall be conducted on the converters submitted for qualification. In addition, the qualifying activity shall have the following options:

(1) the qualification test matrix listed in table III is subject to alteration such that any test may be conducted on any sample submitted for qualification;

(2) inspections may be conducted to verify conformance to any of the requirements that were qualified by similarity.

4.2.3 Qualification retention. Inspection requirements for qualification retention samples shall consist of the examinations and tests listed in table III in accordance with this specification and applicable specification sheet. Samples shall be randomly selected from a production run and submitted to the qualifying activity. Products may be removed from the Qualified Products List (QPL) if:

(1) the manufacturer has not provided the requested certification;

(2) the manufacturer has requested that the product be removed from the list; or

(3) the qualifying activity has removed the product pending the results of requalification testing.

a. The qualifying activity may conduct an evaluation to determine if requalification testing is necessary when any of the following conditions occur:

(1) change in manufacturer's product design, materials, or processes;

(2) change in performance levels or functional requirements due to specification revision;

(3) production facility relocation;

(4) company sale or transfer to new ownership.

4.2.4 Rejection, resubmittal, and retest. Requirements for resubmittal or retest of repaired or reworked test samples will be covered in the contract or purchase order.

4.3 Conformance inspection. Conformance inspection tests shall consist of examinations and tests listed in table IV and the applicable specification sheet. The qualifying activity may conduct conformance inspection on any production components. Unless otherwise directed by the Government contracting officer, delivery of production components should not be delayed pending:

a. results of any test inspection; or
b. correction of failures/defects disclosed during conformance inspection.

Production lot samples will be selected at random under the contract provisions. Production samples forwarded to the qualifying activity will be subjected to conformance inspections in accordance with this specification.

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4.4 Standard test conditions. Unless otherwise specified by the individual tests of 4.6, MIL-STD-810, or the specification sheet, the following standard inspection conditions shall apply:

a. Ambient temperature: 23° ±10 °C (73° ±18 °F).
b. Relative humidity: 50 ±30 percent.
c. Atmospheric pressure: 25.5 to 30.5 inches (648 to 775 mm) of mercury.

4.4.1 Tolerances for the controlled test environment. Unless otherwise specified by the tests of 4.6, the limits for test condition tolerances shall be as follows:

a. Ambient temperature: ±2 °C (±3.6 °F) of the specified value.
b. Air flow: ±5 percent of the specified value.
c. Temperature of forced air: ±2 °C (±3.6 °F) of the specified value.
d. Pressure altitude: ±5 percent of the specified value.
e. Relative humidity: ±5 percent.
f. Vibration Amplitude: Sinusoidal: ±10 percent (+3.0 decibel [dB] -1.5dB)
Random : <500 Hz (-1.5dB)
>500 Hz (+3.0dB)
g. Vibration Frequency: ±2 percent, or ±1/2 Hz below 25 Hz.
h. Acceleration: ±10 percent.

4.4.2 Ambient temperature measurement. When tests are required to be conducted at reduced or elevated temperatures, the ambient temperature shall be determined by measurements of the air at several points surrounding, but not above, the converter and through averaging the results. For converters without fans, air temperature measurements shall be taken as close to the surface of the converter as possible but not so close that the airstream temperature is affected by the converter surface temperature.

4.4.3 Airflow. For all maximum temperature-altitude tests below 20,000 feet (6.1 km), the airflow velocity over the converter within the test chamber shall be adjusted to approximate natural convection conditions and shall not exceed 100 feet (30.5 m)/minute, but shall be adjusted to the minimum velocity consistent with maintaining the required ambient temperature within the specified tolerance. For tests above 20,000 feet (6.1 km) altitude, the airflow may exceed 100 feet (30.5 m)/minute but shall be adjusted to the minimum velocity consistent with maintaining the required ambient temperature within specified tolerance in 4.4.1.

4.4.4 Warm-up and stabilization. Prior to testing, the converter shall be operated, delivering the full rated load specified in the applicable specification sheet for sufficient time for the converter's temperature and output voltage to stabilize. The converter shall be considered to have stabilized when the change over a ten-minute period is measured to be less than:

a. Output voltage: 0.05 volt.
b. Converter critical component temperature: 1.0 °C (1.8 °F).

4.4.5 Voltage measurement. The input and output voltage shall be measured at the external connectors of the converter.

4.4.6 Position. The converter shall be mounted base down.

4.4.7 Electrical loading. Prior to testing, the converter shall be operated at full rated load (see 6.7.7) until stabilized. Unless otherwise specified, the test shall be performed with the

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converter operating at full rated load.

4.4.8 Input voltage. Nominal rated input voltage shall be applied (see 3.5.1.1 (Type AC converter) or 3.5.1.2 (Type DC converter).

4.5 Test equipment and inspection facilities. The inspection facility shall maintain test and measuring equipment capable of meeting the requirements of 4.4.1 to permit the performance of the required inspections and testing.

4.6 Methods of inspection. Input and output voltage, current, and frequency shall be measured during testing in addition to the specific measurement requirements specified by the electrical and operating environment performance inspection methods as applicable.

4.6.1 Examination of product. The converter envelope, geometry, mounting interface, and weight shall be examined for conformance to the applicable specification sheet dimensions and tolerances, as required by 3.4.3 herein and the applicable specification sheet. Visual examinations shall also be conducted to verify that workmanship and construction conform to the specification requirements of 3.4.1 and 3.4.2.

4.6.2 Electrical performance (see 3.5.1, 6.1.4).

4.6.2.1 Voltage regulation. The output voltage shall be measured under the conditions specified in 4.6.2.1.1 through 4.6.2.1.3. For each load, input voltage, position, and temperature-altitude condition, output voltage shall remain within the limits specified in 3.5.1.3.1.1. During the course of the tests, no components of the converter shall exceed their maximum rated temperatures. Tests may be conducted at any intermediate input voltage (see Figure 2 or Figure 4), load, or temperature-altitude condition within the range of Figure 5 at the option of the qualifying activity.

4.6.2.1.1 Voltage regulation at room ambient temperature. Immediately upon energizing the converter (see 3.5.1.1.1.1 (Type AC converter) or 3.5.1.2.1 (Type DC converter), full performance shall be verified by conducting the following tests, a. through e. below. These tests shall be conducted immediately upon energizing the converter and then repeated after operating the converter at nominal input (see 3.5.1.1 (Type AC converter) or 3.5.1.2 (Type DC converter)), and full rated load, until the output voltage is completely stabilized (see 4.4.4).

a. With nominal input voltage specified in 3.5.1.1 (Type AC converter) or 3.5.1.2 (Type DC converter), the load shall be varied from the minimum load (10 percent of rated load) to full rated load in steps of 25 percent rating. When a load between the minimum and 25 percent is required as specified by the applicable specification sheet, the converter shall also be tested at this load. Converters shall be de-energized and re-energized at each load condition to demonstrate their ability to start.

b. The test specified in 4.6.2.1.1a shall be repeated at the maximum input voltage as required by the specification sheet and at 118 VRMS (see 3.5.1.1.1.1a) or 29 VDC (see 3.5.1.2.1a).

c. The test specified in 4.6.2.1.1a shall be repeated at the minimum input voltage as required by the applicable specification sheet and at 108 VRMS (see 3.5.1.1.1.1a (Type AC converter) or 22 VDC (see 3.5.1.2.1a (Type DC converter)).

d. (Applicable to Type AC converters only.) The tests specified in 4.6.2.1.1a, b, c shall be repeated at the maximum input frequency as specified in 3.5.1.1.1.1b (Type AC converter) or in the applicable specification sheet.

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e. (Applicable to Type AC converters only.) The tests specified in 4.6.2.1.1a, b, and c shall be repeated at the minimum input frequency as specified in 3.5.1.1.1.1b (Type AC converter) or in the applicable specification sheet.

4.6.2.1.2 Voltage regulation at maximum temperature-altitude conditions. The converter shall be tested under each of the temperature-altitude conditions listed in table V. Prior to being energized, the converter shall be soaked at the initial maximum temperature, sea level condition for at least one-half hour.

4.6.2.1.3 Voltage regulation at minimum temperature-altitude conditions. The test specified in 4.6.2.1.2 shall be repeated, substituting the temperature-altitude conditions listed in table V and increasing the soak time at the initial -55° C (-67° F) sea level condition to two hours.

4.6.2.2 Worst condition. The converter shall be operated at the worst indicated conditions determined from the tests specified in 4.6.2.1 through 4.6.2.2.

a. Highest output voltage, The conditions from each of worse condition tests resulting in the highest output voltage shall be simultaneously applied.

b. Lowest output voltage. The conditions from each of worse condition tests resulting in the lowest output voltage shall be simultaneously applied.

Prior to the highest output and lowest output voltage tests, the converter shall be soaked at the applicable ambient temperature for a minimum of two hours prior to energizing the converter. The converter shall be operated until voltage, and critical component temperatures stabilize.

4.6.2.3 Efficiency. In conjunction with the test of 4.6.2.1.1, the input and output power of the unit shall be measured. The efficiency shall be calculated under these conditions and shall not be less than specified in 3.5.1.4 under stabilized test conditions as specified in 4.4.4.

4.6.2.4 Input power factor (applicable to Type AC converters only). In conjunction with the tests of 4.6.2.1, the input power factor at half load or greater shall be determined for each condition and shall be as specified in 3.5.1.1.4 (Type AC converter).

4.6.2.5 Output load recovery. The converter shall be tested to verify compliance with 3.5.1.3.2 at maximum, minimum, and room ambient temperatures at sea level altitude. For 100 percent load application and removal, tests shall be conducted at nominal input and maximum input voltages as specified in 3.5.1.1 and 3.5.1.1.1.1 or 3.5.1.2 and 3.5.1.2.1, respectively. For 200 percent load and short circuit application and removal, tests shall be conducted at nominal input voltage, 112.5 VRMS input, and 117.5 VRMS input.

4.6.2.6 Output voltage protection. Protection for failure modes of the converter, which could result in an uncontrolled increase in output voltage, shall be as required by the applicable specification sheet. These failure modes shall be artificially introduced with the converter operating at the maximum input voltage as specified in 3.5.1.1.1.1 (Type AC converter) or 3.5.1.2.1 (Type DC converter) at a minimum and full rated load. The output voltage and output ripple voltage shall not exceed the requirements of 3.5.1.3.3, 3.5.1.3.4, 3.5.1.3.5, and 3.5.1.3.6. Following the test, the converter shall pass the test of 4.6.2.1.1a.

4.6.2.7 Electromagnetic compatibility (see 3.5.1.5). The converter shall be tested in accordance with MIL-STD-461 requirements of CE102, CS101, CS114, CS115, CS116, RE102, and RS103. CE102 shall be performed on converter power output leads as well as power input

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leads, including neutral and returns. Converters intended for use on Army aircraft or Navy antisubmarine aircraft (including land-based aircraft) shall also be in accordance with the requirements of MIL-STD-461: CE101, RE101, and RS101. The test shall be conducted at full rated load and half load with minimum and maximum input voltage. The output voltage shall remain within the normal limits of 3.5.1.3.1.1. During the electromagnetic susceptibility test, the converter shall pass the test of 4.6.3.1. Additionally, the converter shall comply with the bonding requirements of MIL-STD-464. All testing shall be conducted by National Association of Radio and Telecommunications Engineers (NARTE) certified personnel at a National Voluntary Accreditation Program (NVLAP) accredited laboratory.

4.6.2.8 Input power susceptibility (see 3.5.1.1.2 (Type AC converter), 3.5.1.2.3 (Type DC converter).

4.6.2.8.1 Abnormal steady-state input voltage. The converter shall be operated at the no load and the impedance equivalent of full rated load at 5 VDC output with input voltages ranging from 100 to 125 VRMS in 5 volt steps (see 3.5.1.1.1.2a and b (Type AC converter)) or input voltages 20 to 31.5 VDC in 3 volt steps (see 3.5.1.2.1a and b (Type DC converter)). The converter shall be operated for five minutes under each input voltage/load combination and shall meet the requirements of 3.5.1.1.1.2 (Type AC converter) or 3.5.1.2.2 (Type DC converter). Following this test, the converter shall pass the test of 4.6.3.1.

4.6.2.8.2 Input waveform distortion (Type AC converter only) (see 3.5.1.1.1.1c). The test of 4.6.2.1.1a shall be conducted at half- and full-rated load under the following conditions:

a. Using an AC power source that normally provides 3.0 percent total distortion or less while supplying the converter at full rated load, sufficient input inductance shall be added equally in each line (except the neutral line if utilized) between the power source and the converter to increase input waveform distortion to the maximum allowed by 3.5.1.1.1.1c (Type AC converter). During this test, the output voltage shall meet the requirements of 3.5.1.3.1.1. Input inductance shall then be further increased to obtain the maximum distortion allowed by 3.5.1.1.1.2c (Type AC converter). During this test, output voltage shall meet the requirements of 3.5.1.3.1.1.

b. Using a variable waveform AC power source, the input waveform distortion shall be adjusted to the maximum allowed by 3.5.1.1.1.1c (Type AC converter) while maintaining a crest factor (see 6.7.4) of 1.31 to 1.40. This test shall be repeated while maintaining a crest factor of 1.42 to 1.51. During these tests, output voltage shall meet the requirements of 3.5.1.3.1.1. Distortion shall then be adjusted to the maximum allowed by 3.5.1.1.1.2c (Type AC converter) while maintaining a crest factor of 1.26 to 1.38. This test shall be repeated while maintaining a crest factor of 1.44 to 1.56. During these tests, output voltage shall meet the requirements of 3.5.1.3.1.2.

4.6.2.8.3 Transient input voltage.

4.6.2.8.3.1 Transient surge susceptibility.

4.6.2.8.3.1.1 Full performance. With the converter operating at minimum load (10 percent of rated load) and full rated load with nominal input, the positive step voltages indicated in table VI (Type AC converter) or table VII (Type DC converter) shall be applied to the converter input terminals four times for each condition (see 3.5.1.1 (Type AC converter) or 3.5.1.2 (Type DC converter)). The converter shall meet the requirements of 3.5.1.1.2.1.1 (Type AC converter) or

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3.5.1.2.3.1.1 (Type DC converter). Following the test, the converter shall pass the test of 4.6.3.1.

4.6.2.8.3.1.2 Abnormal and emergency operation (see 3.5.1.1.2.1.2, 3.5.1.2.2, 3.5.1.2.3.1.2). With the converter operating at minimum load (10 percent of rated load) and full rated load with nominal input, the positive step voltages indicated in table I (Type AC converter) or table II (Type DC converter) shall be applied to the converter input terminals four times for each condition (see 3.5.1.1 (Type AC converter) or 3.5.1.2 (Type DC converter)). The converter shall be observed for evidence of flashover or insulation breakdown and shall meet the requirement of 3.5.1.1.2.1.2 (Type AC converter) or 3.5.1.2.3.1.2 (Type DC converter). Following the test, the converter shall pass the test of 4.6.3.1.

4.6.2.8.4 Transient input frequency (Type AC converter only). The transient input frequencies of 3.5.1.1.3 (Type AC converter) shall be applied to the input of the converter using a transient generator. Following the test, the converter shall pass the test of 4.6.3.1.

4.6.3 Operating environment performance.

4.6.3.1 Performance verification. Unless otherwise specified, satisfactory converter operation shall be verified during and after the following environmental tests. The tests of 4.6.2.1.1a shall be conducted to verify converter performance, except that both positive and negative peak to mean ripple voltages shall also be measured at each condition. Output voltage shall meet the requirements of 3.5.1.3.1.1.

4.6.3.2 Acceleration (see 3.5.2.2). The converter shall be tested in accordance with the acceleration test of MIL-STD-810, Method 513, Procedure II. The required acceleration level for all directions in procedure II (operational test) shall be 6.0g and shall be applied along three mutually perpendicular axes in two opposite directions along each axis. The test time duration in each direction shall be at least one minute following centrifuge stabilization, and the item shall be operated continuously at full rated load and nominal input (see 3.5.1.1 (Type AC converter) or 3.5.1.2 (Type DC converter)). The output voltage shall be monitored during the test to determine compliance with 3.5.1.2.3.1.1. At the conclusion of the test, the converter shall be examined for evidence of damage and shall pass the test of 4.6.3.1.

4.6.3.3 Shock (see 3.5.2.3). The converter shall be subjected to shock testing in accordance with those methods described in 4.6.3.3.1 and 4.6.3.3.2. During shock testing, the converter shall be oriented as shown on Figure 6 using the mounting provisions specified in the applicable specification sheet.

4.6.3.3.1 Functional (see 3.5.2.3a). The converter shall be operating during the functional shock test at full rated load and nominal input (see 3.5.1.1 (Type AC converter) or 3.5.1.2 (Type DC converter)). The output voltage shall be monitored during the test to determine compliance with 3.5.1.2.3.1.1. The applied shock response spectrum (SRS) testing per MIL-STD-810, Method 516 Procedure I (functional shock) using the reference profile shown on figure 9. The converter shall be subjected to three shocks in both directions along each of three mutually perpendicular axes for a total of 18 shocks at the functional level. At the conclusion of the functional shock test, the converter shall be inspected for evidence of damage and shall pass the test of 4.6.3.1.

4.6.3.3.2 Crash hazard (see 3.5.2.3b). The converter shall be subjected to shock response spectrum (SRS) testing per MIL-STD-810, Method 516 Procedure V (crash hazard shock) using the reference profile shown on figure 9. The converter shall be subjected to two shocks in each

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direction along each of three mutually perpendicular axes for a total of 12 shocks at the crash hazard level. The converter is not required to demonstrate specific performance during the crash hazard test; however, it shall remain securely attached to the fixture and pose no missile hazard to personnel or equipment. Bending and distortion of the converter shall be permitted.

4.6.3.4 Vibration. The converter shall withstand the vibration environment induced by aircraft operations, as simulated by the vibration tests herein. Vibration testing shall be performed according to the general vibration guidelines of MIL-STD-810, Method 514, Procedure I (general vibration), or the profile specified in the applicable specification sheet for a particular aircraft application. Sinusoidal dwell vibration shall be applied along each of three mutually perpendicular axes of the converter. The input levels specified by the vibration reference profile specified in 4.6.3.4.2 shall be maintained at the equipment mounting locations. Upon completion of the vibration tests, component electrical performance shall be within specified limits. All component parts, connections, and mounting hardware shall exhibit no evidence of cracking, rupture, or operational degradation, and the converter shall meet the requirements of 3.5.2.4.

4.6.3.4.1 Resonance search. Resonant frequencies of the equipment shall be determined by performing a sinusoidal sweep from 5 - 2000 Hz. The time for sweep (5 Hz to 2000 Hz to 5 Hz) shall be 20 minutes. The applicable reference profile specified in 4.6.3.4.2 shall be used. A resonance shall be defined as a frequency at which the response of the test article is greater than or equal to two times the input vibration magnitude.

4.6.3.4.2 Sinusoidal vibration test. The converter shall be subjected to 3 hours of sinusoidal vibration testing comprising both dwell testing and cycling. The reference profile shall be defined as follows: 5-2000 Hz, 0.036 inch double amplitude, 10 g maximum (see Figure 7). Sinusoidal dwells shall be performed for 30 minutes at each of the four highest resonance frequencies. Sinusoidal cycling testing at a sweep rate of 1.0 octave/minute shall be performed for 1.0 hour. If less than four resonant frequencies are identified, the allocated dwell time shall be added to the cycling portion of the test. For equipment weighing 80 lbs. (36 kg) or more, the maximum input vibration level may be reduced by .035 oz (1.0 g) for every 20 lbs. (9 kg) increment over 80 lbs. (36 kg). Following the vibration test, the converter shall pass the test of 4.6.3.1 and shall be thoroughly inspected for cracking or rupture of any components, solder connections, or of the mounting assembly itself. The presence of such damage shall be cause for rejection.

4.6.3.5 Fungus. Fungus testing is not required if all the parts and materials are certified as non-nutrient supporting. A certificate signed by an individual authorized to speak for the manufacturer may be submitted in lieu of fungus testing with the approval of the qualifying activity. The certificate shall state that the converter is constructed of non-nutrient materials and does not use any organic material (see 6.2.d (8)). If a signed certificate is not available, the converter and installation instructions shall be tested in accordance with MIL-STD-810, test method 508, to determine compliance with 3.5.2.5. At the conclusion of the test, the converter shall be removed from the test chamber, and excess moisture may be removed by turning the test item upside down or by shaking. No washing or wiping of the test item is permitted. The converter shall be inspected for fungus growth and shall pass the test of 4.6.3.1.

4.6.3.6 Salt fog. The converter and mounting hardware shall be tested in accordance with MIL-STD-810, test method 509, to determine compliance with 3.5.2.6. The converter shall be oriented base down. The duration of the exposure shall be 48-hours and 48 hours of drying with

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the converter non-operating. The test cycle shall consist of 24 hours of a 5% +/- 1% salt solution concentration exposure followed by 24 hours of drying. Two test cycles shall be conducted. At the completion of the second 24 hours of drying, a functional test shall be conducted, and the converter shall pass the test of 4.6.3.1. The converter shall then be physically inspected both inside and out. There shall be no evidence of (1) material degradation such as corrosion; or (2) blistering of paint or degradation of electrical insulation. Metal parts shall be examined for corrosion, moving parts for clogging or binding and paint for blistering.

4.6.3.7 Explosive atmosphere (see 3.5.2.7). The converter shall be tested in accordance with MIL-STD-810, test method 511, Explosion atmosphere - Procedure I. The test altitude shall be 40,000 feet (12.2 km) or the high altitude specified in the applicable specification sheet. The test temperature shall be 71° C (160° F) or the high temperature at sea level specified in the applicable specification sheet.

4.6.3.8 Contamination by fluids. Determine conformance to the requirements of 3.3.4 by test or by similarity (see 6.2d (7)). The converter shall be tested in accordance with MIL-STD-810, test method 504, Procedure I (see 6.8) for additional fluids if not listed in method 504).

4.6.3.9 Toxicity and fire resistance. Determine conformance of component materials to the requirements of 3.3.5 by test or by similarity (see 6.2d (7)). If testing is required, components shall be subjected to a flammability test in accordance with MIL-STD-202-111.

4.6.3.10 Electrostatic discharge. The electrostatic discharge requirement of 3.5.2.8 shall be verified by the electrostatic discharge test of section 25 of RTCA/DO-160. Test points shall be the top of the converter case and the outside of each connector.

4.6.4 Disassembly and inspection. At the conclusion of the test program, each sample shall be disassembled, as necessary, to inspect for evidence of deleterious effects of qualification testing; examples of these effects include, but are not limited to, wear, defects, and overheating. The presence of such evidence is cause for rejection.

4.6.5 Burn-in (see 3.9). The converter shall be subjected to one or more cycles of testing conforming to Figure 8. The number of cycles shall be determined by referencing table VIII. The heating and cooling temperature ramps on Figure 8 shall be performed at a minimum of 5° C (41° F) /minute. The converter shall be turned off during the cool-down period (cooling ramp) and turned on for the balance of each cycle. The equipment shall be vibrated in one axis with an amplitude of $0.02g^2$/Hz over a frequency range of 20 - 2,000 Hz during the last 20 minutes of each four-hour stabilized temperature condition. Failures may occur and be corrected in any cycle prior to the final cycle. The burn-in is considered acceptable if there is no failure during the final cycle. If a failure occurs in the final cycle, the equipment failure may be corrected, subject to approval by the qualifying agency, and a failure-free cycle run.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing

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Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The items described herein are intended for use on aircraft to convert single-phase, 115 VRMS, 400 Hz AC or 28 volts DC power to 5 VDC power. Converters typically are used to provide 5 VDC power on aircraft with main electrical generating systems that provide 115 VRMS, 400 Hz AC or 28 Volts DC power. The military operating environment and electrical performance requirements of this specification exceed the requirements used for commercial aircraft converters. The converter operates strictly as a power supply. Peripheral equipment battery protection should be provided by the peripheral equipment being charged. The converter is not required to operate during Abnormal and Emergency operations and is not intended to be used to charge flight or safety-critical equipment. Converter design should also consider use on aircraft compliant to MIL-STD-704 variable frequency (360 to 800 Hz).

6.1.1 Installation. The 5 VDC power is typically supplied to a USB connector to charge peripheral equipment and does not have data exchange capability. Unless otherwise specified in the applicable specification sheet, the output power should be routed to a USB-IF 3.1 compliant USB Type-A connector. Power from the converter to the aircraft charging connector should be routed using aircraft wiring size 20 to 24 depending on the charging current requirements. Twisting of the aircraft wiring is not required because there is no digital data being routed. If the aircraft charging connector is not a USB Type-A connector, a non-USB cable or adapter may be required to connect the peripheral equipment being charged. Charging voltage is conducted through the USB pin 1 - +5VDC and pin 4 - ground of the peripheral equipment. A maximum of 200-ohm resistance should be connected to the USB pins 2 and 3. This resistance may be connected at the converter output or the aircraft charging connector. The purpose of the resistance on pins 2 and 3 is to identify the converter as a Dedicated Charging Port (see BC 1.2, Copies of BC 1.2 are available online from the USB Enabling Connections website at https://www.usb.org/documents/).

6.1.2 Peripheral equipment Information Technology (IT) compatibility (see 3.9.8). Peripheral equipment charged by the converter should be tested and comply with DoD cybersecurity policies as specified in DoDI 8500.01, which will allow the system to receive an authorization to operate (ATO) based upon the Risk Management Framework (RMF) process as specified in DoDI 8510.01. Peripheral equipment should be designed to securely operate without compromised confidentiality, integrity, or availability. Copies of DoDI 8500.01 and DoDI 8510.01 are available online at https://dodcio.defense.gov/Library/.

6.1.3 Peripheral equipment lithium battery. Peripheral equipment using lithium batteries should comply with the Navy Lithium Battery Safety Program (NAVSEA 9310.1C). Lithium batteries should be in accordance with NAVSEA S9310-AQ-SAF-010. Copies of NAVSEA S9310-AQ-SAF-010 are available online at https://navysbir.com/n21_1/Topic-N211-033-Reference_Document_S9310-AQ-SAF-010-Rev3.pdf.

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6.1.4 MIL-HDBK-704 test procedures. MIL-STD-704 states that equipment testing is required to demonstrate that utilization equipment is compatible with the electric power characteristics specified in MIL-STD-704. MIL-STD-704 is the interface document that defines the aircraft electrical power characteristics at the input terminals to utilization equipment. The aircraft electrical system is required to provide power in accordance with MIL-STD-704, and the utilization equipment is required to perform as specified when provided input power in accordance with MIL-STD-704. MIL-HDBK-704-1 through MIL-HDBK-704-8 provide detailed test methods for demonstrating compliance to MIL-STD-704. MIL-HDBK-704-2 provides guidance for single-phase, 400 Hz, 115-volt utilization equipment, and MIL-HDBK-704-8 provides guidance for 28 VDC utilization equipment. See NAVAIRINST 3080.1 in regard to policy and responsibility for compliance and testing to MIL-STD-704. Copies of NAVAIRINST 3080.1 are available online at https://myteam.navair.navy.mil/km/71/Directives/Directives_Instructions/NAVAIRINST%203080.1C.pdf.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Military part number.
c. Packaging (see 5.1).
d. Contract provisions for acquisition of manufacturer's data may include:
   (1) Schedules for component design, development, and delivery;
   (2) Wiring diagrams, schematics, outline drawings;
   (3) List of parts, list of materials, functional descriptions, operating instructions,
   (4) Installation instructions and servicing requirements;
   (5) Temperature, altitude, and air flow rate criteria correlated with converter output capacity;
   (6) Contractor analyses of sneak circuits, failure modes and effects, parts stress, parts failure rate modeling, and reliability prediction;
   (7) Stress screening test procedures, acceptance test procedures, manufacturer's test data, and qualification by similarity data;
   (8) Verification data, manufacturing process controls, inspection criteria, configuration management data, engineering change processes, and parts traceability.
   (9) Re-inspection date marking and any special marking.
   (10) Handbooks or manuals, when required.
   (11) Contamination of fluids analysis.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List (QPL)-32729 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be

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obtained from the Naval Air Warfare Center – Aircraft Division (AB43200), 48298 Shaw Road, Patuxent River, MD 20670-1900. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Provisions governing qualification. Qualification will be conducted in accordance with the provisions and guidelines of SD-6, "Provisions Governing Qualification." Copies of this document are available online at https://quicksearch.dla.mil.

6.3.2 Qualification sample submittal. Qualification submissions should be submitted to the qualifying activity. The qualifying activity may acquire up to four converters to conduct qualification tests. Each converter will be acquired with installation hardware and a mating connector plug for each receptacle.

6.4 Standard design practices. The following paragraphs contain design guidelines based on actual experience that should be considered by the manufacturer when a converter is to be developed to meet the performance requirements of this specification.

6.4.1 Verification and configuration management. As required by the contract, verification and failure analysis/reporting should be in accordance with industry standards, such as ISO 9000. Product identification, parts traceability, product configuration control, and documentation data control should be in accordance with industry standards for configuration management, such as SAE EIA-649 or ISO 10007. MIL-HDBK-61 also provides information on configuration management. Copies of ISO 9000, ISO 10007, and SAE EIA-649 are available from https://webstore.ansi.org or https://www.techamerica.org. Copies of MIL-HDBK-61 are available from https://quicksearch.dla.mil/.

6.4.2 Materials and structural design. Military standard parts are desired. Commercial industry parts, such as screws, bolts, nuts, and washers, may be used provided that the performance requirements of this specification are met under all specified operating conditions and environments. Design characteristics and limitations should be observed for the materials and components selected for use in the converter.

6.4.2.1 Recycled, recovered, or environmentally preferable materials. The use of toxic chemicals, hazardous substances, and ozone-depleting chemicals should be avoided in the preparation and production of equipment. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements specified herein and promotes economically advantageous life cycle costs. Information regarding toxic chemicals, hazardous substances, and ozone-depleting substances can be found at the Environmental Protection Agency website at https://www.epa.gov.

6.4.2.2 Fasteners, inserts, and retention. Screws, fasteners, inserts should be selected using industry-standard NASM1515 for guidance.

6.4.2.3 Corrosion. Corrosion resistance treatments for aluminum alloy materials are provided in MIL-PRF-8625. Corrosion protection methods for magnesium alloy materials are provided in SAE AMS-M-3171.

6.4.3 Electrical and electronics design.

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6.4.3.1 Electronics internal wiring and electric connections. Printed circuit board design and performance criteria guidance is provided in MIL-PRF-31032, MIL-STD-1547, MIL-HDBK-1861, and industry standards IPC-2221. Guidelines 1, 10, 17, 19, 20, and 69 of MIL-HDBK-454 provide guidance on Safety Design Criteria – Personnel Hazards, Electrical Connectors, Printed Wiring, Internal Wire Hookup, and Internal Wiring Practices. Copies of IPC 2221 are available from https://www.ipc.org/.

6.4.3.2 Semiconductors, resistors, and capacitors. MIL-STD-750 and MIL-PRF-19500 provide guidance on the design and test of semiconductor devices. Adjustable resistors should conform to MIL-PRF-39015. MIL-HDBK-198 provides guidance on capacitor selection. MIL-HDBK-199 provides guidance on resistor selection. Silver case wet slug tantalum capacitors should not be used.

6.4.3.3 Microelectronic devices. Microelectronic devices should be selected in accordance with Guideline 64 of MIL-HDBK-454. Microelectronic devices qualified to MIL-STD-883, and listed in MIL-PRF-38535, are preferred. Static discharge control should be in accordance with ANSI/ESD S20.20 or JESD625. Copies of ANSI/ESD S20.20 are available from https://esda.org and copies of JESD625 are available from https://www.jedec.org.

6.4.3.4 Parts de-rating. DoD standard practices for electrical and electronic parts de-rating are documented in SD-18 available at https://www.navsea.navy.mil/Home/Warfare-Centers/NSWC-Crane/Resources/SD-18/.

6.4.4 Reliability. Reliability should be considered in each phase of the converter design process. MIL-STD-790, MIL-HDBK-217, and MIL-HDBK-251 provide guidance on established reliability and reliability prediction.

6.4.5 Maintainability. The converter should be designed to facilitate assembly, disassembly, location of trouble sources, and maintenance without the aid of special tools. All functional parts should be readily identifiable, accessible, and removable for replacement. When printed circuit boards are utilized, only clear transparent conformal coating should be used in order to permit identification, location, removal, and replacement of functional parts.

6.4.6 Identification markings. MIL-STD-130, MIL-STD-1285, and MIL-STD-1812 provide guidelines for the identification marking of equipment and parts.

6.4.7 Calibration system. ISO/IEC 17025 and ISO 10012 provide guidance for the establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment.

6.4.8 Workmanship. Guideline 9 of MIL-HDBK-454 provides guidance on workmanship practices.

6.4.9 Installation. The aircraft manufacturer should allow clearance of at least one inch at each end of the converter in addition to the maximum dimensions shown on the applicable specification sheet for installation and ventilation.

6.5 Vibration. Converters meeting the vibration requirements specified herein are suitable for use in jet, propeller, and helicopter applications. Specific aircraft platform requirements specified in the applicable specification sheet may limit the suitable use to the specific platform.

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6.6 Packaging. The converters should be packaged in accordance with MIL-STD-2073-1 level A or B, as specified in the contract. Interior and exterior containers should be marked in accordance with MIL-STD-2073-1 (see 5.1).

6.7 Definitions. The definitions in MIL-STD-704, MIL-STD-810, and MIL-STD-461 will apply in addition to those cited herein. Normal operation and abnormal operation characteristics for aircraft electrical systems are specified in MIL-STD-704, including definitions of transient and steady-state conditions for voltage and frequency.

6.7.1 Abnormal input voltage range. The abnormal input voltage range is the voltage range encountered during operation from aircraft batteries or other emergency power sources. Abnormal input voltage occurs when a malfunction or failure in the aircraft electrical system has taken place, and the protective devices of the system are operating to remove the malfunction or failure from the remainder of the system before the limits for abnormal operation are exceeded. The aircraft power source may operate in a degraded mode on a continuous basis where the power characteristics supplied to the aircraft utilization equipment exceed normal operation limits but remain within the limits for abnormal operation.

6.7.2 Alternating current. Alternating currents used in this specification are root mean square (RMS) values (see MIL-STD-704).

6.7.3 Alternating voltage. Alternating voltages used in this specification are considered line-to-neutral RMS values for both 3 wire and 4 wire input connections (see MIL-STD-704).

6.7.4 Crest factor. The crest factor is the absolute value of the ratio of the peak to the RMS value for each half cycle of the voltage waveform.

6.7.5 Direct voltage and current. Direct voltages and currents used in this specification are mean (average) values (see MIL-STD-704).

6.7.6 Emergency operation. Emergency operation occurs following the loss of the aircraft main generating equipment when a limited electric source, independent of the main system, is used to power a reduced complement of distribution and utilization equipment selected to maintain flight and personnel safety.

6.7.7 Full rated load (full load). The full rated load used in this specification is the maximum continuous converter output load specified in the applicable specification sheet for a given temperature-altitude condition.

6.7.8 Normal input voltage range. The normal input voltage range is the voltage range encountered when all aircraft engines are operating and the aircraft electrical system is fully operational. Normal input voltage occurs when the aircraft electrical system is operating as intended in the absence of any fault or malfunction that degrades performance beyond established requirements. It includes all system functions required for aircraft operation except during the electric starting of propulsion engines and the battery start of an auxiliary power unit. Normal input voltage includes aircraft switching of utilization equipment, prime mover speed changes, synchronizing and paralleling of power sources, and operation from external power sources.

6.8 Aviation fuel and hydraulic fluids. Testing for resistance to the effects of oil, fuel, and hydraulic fluids should also include those listed in MIL-PRF-23699, MIL-PRF-7808,

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DoD-PRF-85734, and MIL-DTL-5624 (JP-4/JP-5), MIL-PRF-8188, MIL-PRF-5606, and MIL-PRF-87257.

6.9 Aluminum alloy castings. Defects not materially affecting the suitability of the castings may be repaired at the foundry or during machining by peening, impregnating, welding, or other methods acceptable to the qualifying activity.

6.10 Specification sheet requirements. Specification sheets define the performance, design, interface, test, qualification, etc., requirements for a specific converter and its components. The paragraphs and tables in this specification identified in table IX require information to be provided in the specification sheet and provide the minimum requirements to define the specific converter. The following applies: Paragraphs with an X in the "As Required By" column will either have a default value given in this specification or must have a value or requirement given in the specification sheet. Paragraphs with an X in the "Unless Otherwise Specified" column will have the requirement or value given only in the specification sheet, if it is required.

6.11 Electromagnetic compatibility. Aircraft applications that have a concern for lightning protection should consider the requirements of MIL-STD-461 CS117 (Lightning).

6.12 Subject term (key word) listing.

AC supply
Electrical
Single-phase
USB

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TABLE I. Input - Type AC converter abnormal and emergency operation transient surge susceptibility (see 3.5.1.1.2.1.2, 4.6.2.8.3.1.2).

VRMS	Duration (seconds)
180	0.010
162	0.065
124	25.00
102	25.00
64	0.010
5	0.050
0	0.050

TABLE II. Input - Type DC abnormal and emergency operation transient surge susceptibility. (see 3.5.1.2.3.1.2, 4.6.2.8.3.1.2).

VDC	Duration (seconds)
50	0.010
43	0.080
30	25.00
21	25.00
15	0.010
5	0.050
0	0.050

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TABLE III. Qualification inspections (see 4.2.1, 4.2.2a, 4.2.2a(1) and 4.2.3).

Inspection title (Examination or test)	Type AC Requirement paragraph	Type DC Requirement paragraph	Test method paragraph	Qualification (sample no.)			
				1	2	3	4
Examination of product	3.3, 3.4, 3.7, 3.8	3.3, 3.4, 3.7, 3.8	4.6.1	X	X	X	X
Voltage and frequency regulation	3.5.1.3.1.1, 3.5.1.3.1.2	3.5.1.3.1.1, 3.5.1.3.1.2	4.6.2.1				
At room ambient temperature conditions	---	---	4.6.2.1.1	X	X	X	X
At maximum temperature-altitude conditions	---	---	4.6.2.1.2	X	X		
At minimum temperature-altitude conditions	---	---	4.6.2.1.3	X	X		
Worst condition	---	---	4.6.2.2	X	X		
Efficiency	3.5.1.4	3.5.1.4	4.6.2.3	X	X	X	X
Input power factor	3.5.1.1.4	---	4.6.2.4	X	X	X	X
Output load recovery	3.5.1.3.2	3.5.1.3.2	4.6.2.5	X		X	
Output voltage protection	3.5.1.3.3, 3.5.1.3.4, 3.5.1.3.5, 3.5.1.3.6	3.5.1.3.3, 3.5.1.3.4, 3.5.1.3.5, 3.5.1.3.6	4.6.2.6	X		X	
Input power susceptibility	3.5.1.1.2	3.5.1.2.3	4.6.2.8	X		X	
Electromagnetic compatibility	3.5.1.5	3.5.1.5	4.6.2.7	X		X	
Input Waveform distortion	3.5.1.1.1.1c	---	4.6.2.8.2	X	X		
Acceleration	3.5.2.2	3.5.2.2	4.6.3.2	X			
Shock	3.5.2.3	3.5.2.3	4.6.3.3	X			
Vibration	3.5.2.4	3.5.2.4	4.6.3.4		X		X
Fungus	3.5.2.5	3.5.2.5	4.6.3.5				X
Salt fog	3.5.2.6	3.5.2.6	4.6.3.6		X		
Explosive atmosphere	3.5.2.7	3.5.2.7	4.6.3.7		X		
Electrostatic discharge	3.5.2.8	3.5.2.8	4.6.3.10			X	X
Contamination by fluids	3.3.4	3.3.4	4.6.3.8			X	
Toxicity and fire resistant	3.3.5	3.3.5	4.6.3.9			X	
Disassembly and inspection	---	---	4.6.4			X	X

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TABLE IV. Conformance inspections (see 4.3).

Inspection title (Examination or test)	Requirement paragraph	Test method paragraph
Examination of product	3.3, 3.4, 3.7, 3.8	4.6.1
Burn-in	3.9	4.6.5
Voltage regulation at room ambient temperature	---	4.6.2.1.1
Efficiency	3.5.1.4	4.6.2.3
Input power factor	3.5.1.1.4	4.6.2.4

TABLE V. Temperature-altitude conditions (see 4.6.2.1.2, 4.6.2.1.3).

Minimum		Maximum	
Ambient temperature °C (°F)	Altitude Feet (km)	Ambient temperature °C (°F)	Altitude Feet (km)
-55 (-67)	Sea level	71 (160)	Sea level
	15,000 (4.6)	25 (77)	15,000 (4.6)
	25,000 (7.6)	8 (46)	25,000 (7.6)
	35,000 (10.7)	-10 (14)	35,000 (10.7)
	50,000 (15.2)	-10 (14)	50,000 (15.2)

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TABLE VI. Input - Type AC converter full performance transient surge susceptibility (see 4.6.2.8.3.1.1).

VRMS	Duration (seconds)
180	0.010
165	0.029
145	0.054
130	0.073
95	0.048
80	0.010

TABLE VII. Input - Type DC full performance transient surge susceptibility (see 4.6.2.8.3.1.1).

VDC	Duration (seconds)
50	0.010
40	0.050
30	15
20	20

TABLE VIII. Number of burn-in cycles (see 4.6.5).

Number of electronic parts	Number of cycles
0 – 100	1
101 – 500	3
501 – 2000	6
2001+	10

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TABLE IX. Specification sheet requirements (see 6.10).

Paragraph	As required by	Unless otherwise specified
3.3.7 Color		X
3.4.2.1.1 External electric connections	X	
3.4.2.2 Electrical power routing		X
3.4.3.5 Cooling	X	
3.4.3.6 Operating position		X
3.4.3.7 Size and weight	X	
3.5.1 Electrical	X	
3.5.1.1.1.1.a Type AC Full performance steady state input voltage		X
3.5.1.1.1.1.b Type AC abnormal and emergency operation steady-state input frequency		X
3.5.1.1.1.2.a Type AC abnormal and emergency operation Steady state input voltage between 100 VRMS and 108 VRMS		X
3.5.1.1.1.2.b Type AC abnormal and emergency operation Steady state input voltage between 118 VRMS and 125 VRMS		X
3.5.1.1.1.2.c Type AC abnormal and emergency operation Non-harmonic frequency components		X
3.5.1.2.1a Type DC full performance steady-state input voltage		X
3.5.1.2.2.a Type DC Abnormal and emergency operation steady-state input between 20 VDC and 22 VDC voltage		X
3.5.1.2.2.b Type DC Abnormal and emergency operation steady-state input between 22 VDC and 31.5 VDC		X
3.5.1.2.2.c Type DC Abnormal and emergency operation distortion factor		X
3.5.1.4 Efficiency	X	
3.6 Reliability		X
3.7.2 Mechanical nameplate	X	
3.7.3 Electrical connections	X	
4.4 Standard test conditions		X
4.4.4 Warm-up and stabilization	X	
4.4.7 Electrical loading		X
4.6.1 Examination of product	X	
4.6.2.1.1.a Nominal input voltage regulation at room ambient temperature	X	
4.6.2.1.1.b Maximum input voltage regulation at room ambient temperature	X	

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TABLE IX. Specification sheet requirements (see 6.10) - Continued.

Paragraph	As required by	Unless otherwise specified
4.6.2.1.1.c Minimum input voltage regulation at room ambient temperature	X	
4.6.2.1.1.d Maximum input frequency regulation at room ambient temperature	X	
4.6.2.1.1.e Minimum input frequency regulation at room ambient temperature	X	
4.6.2.6 Output overvoltage protection	X	
4.6.3.1 Performance verification		X

TABLE X. Parts de-rating (see 3.4.2.3).

Component type	Operating parameter	De-rate to (e.g., percent of rating)
Resistors	Power	50%
	Voltage	80%
Capacitors	Voltage	60%
	Ripple voltage	50%
	Ripple current	70%
Inductors and transformers	Hot spot temperature	30 °C below rating of insulation
Semiconductors	Junction temperature	20 °C below maximum rated temperature
	Power	70%
	Current	70%
	Voltage	70% 1/

1/ 100% for integrated circuit components designed to operate at a specific regulated DC voltage.

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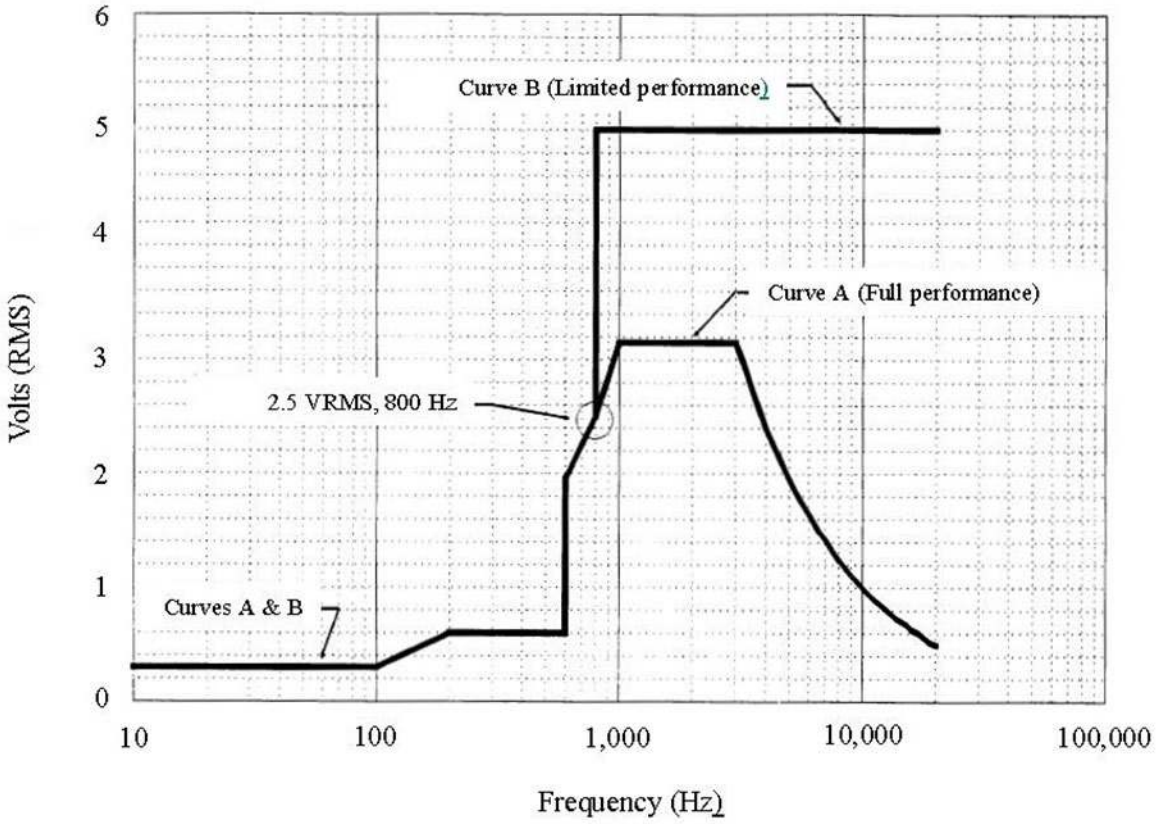


FIGURE 1. Input - Type AC input waveform distortion (see 3.5.1.1.1.2.c).

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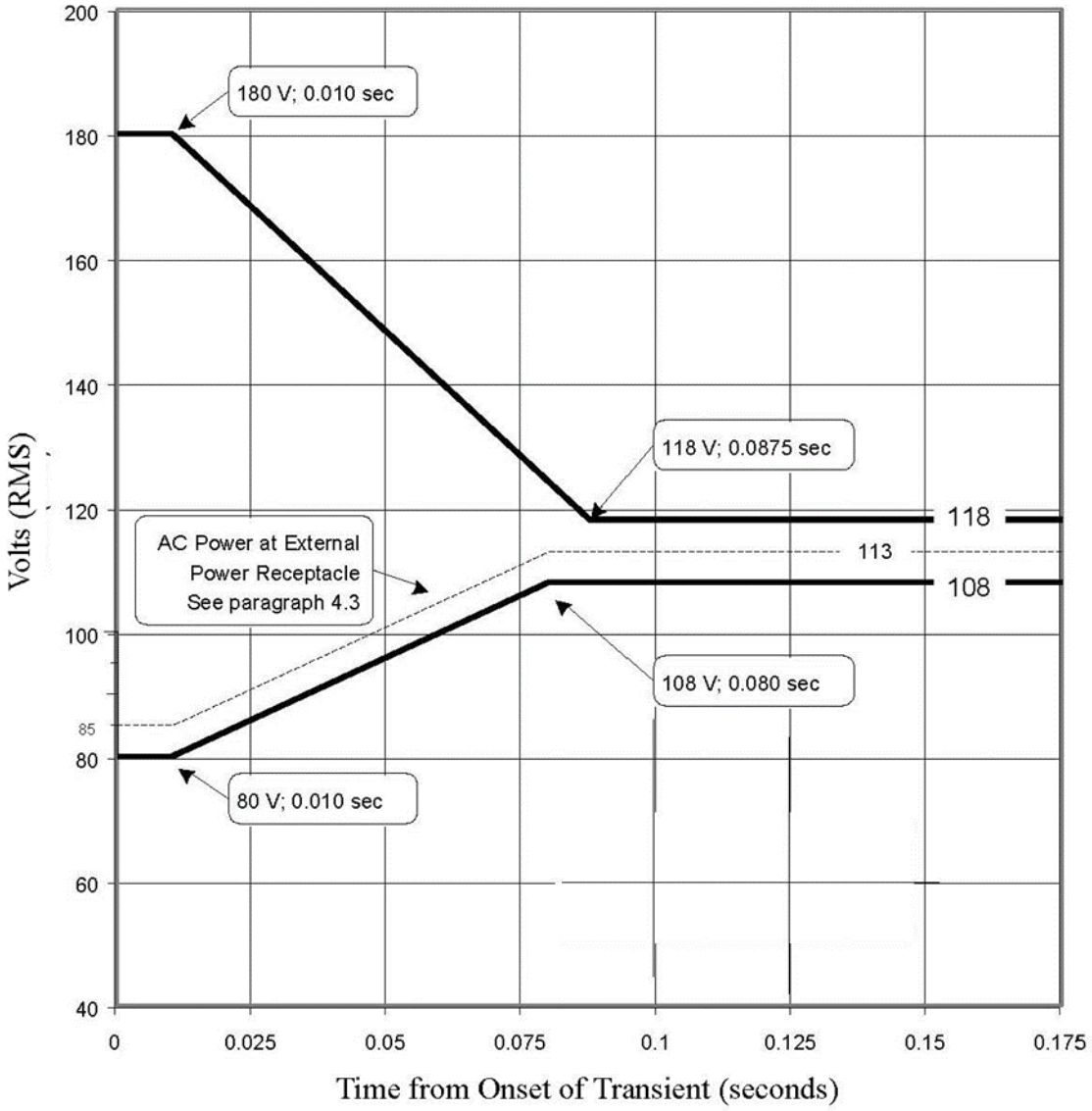


FIGURE 2. Input - Type AC converter 400 Hz voltage transient limits (see 3.5.1.1.2.1.1, 4.6.2.1).

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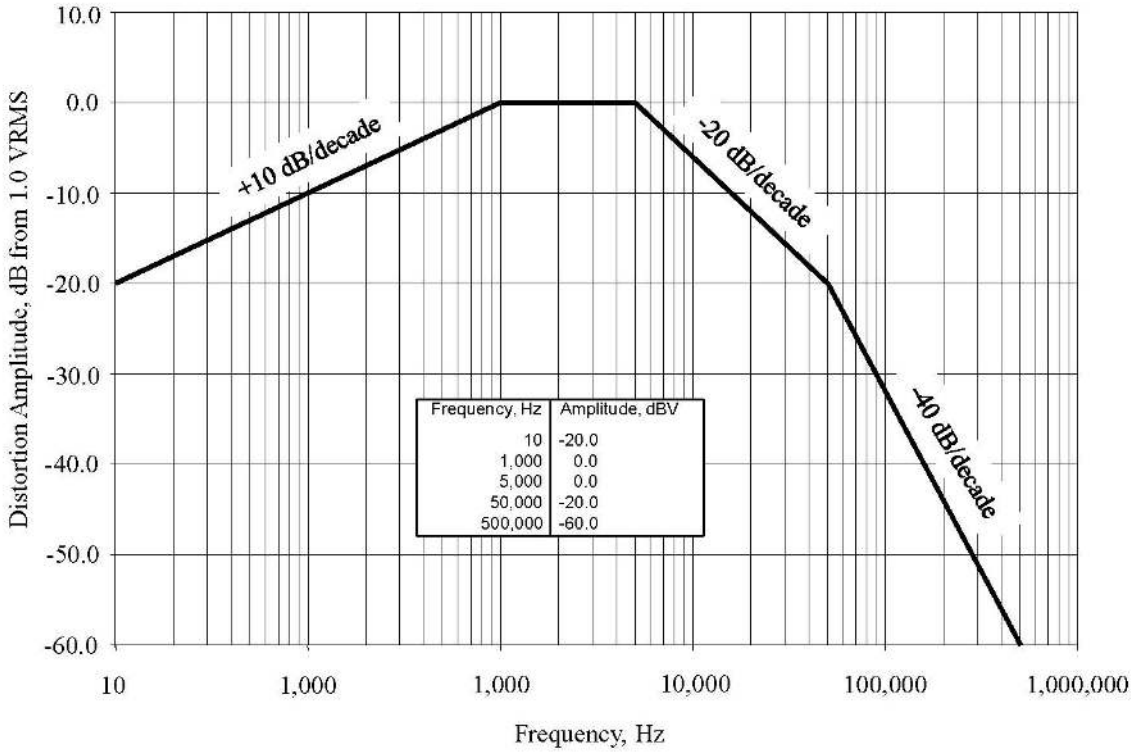


FIGURE 3. Input - Type DC converter input maximum distortion spectrum (see 3.5.1.2.2.c).

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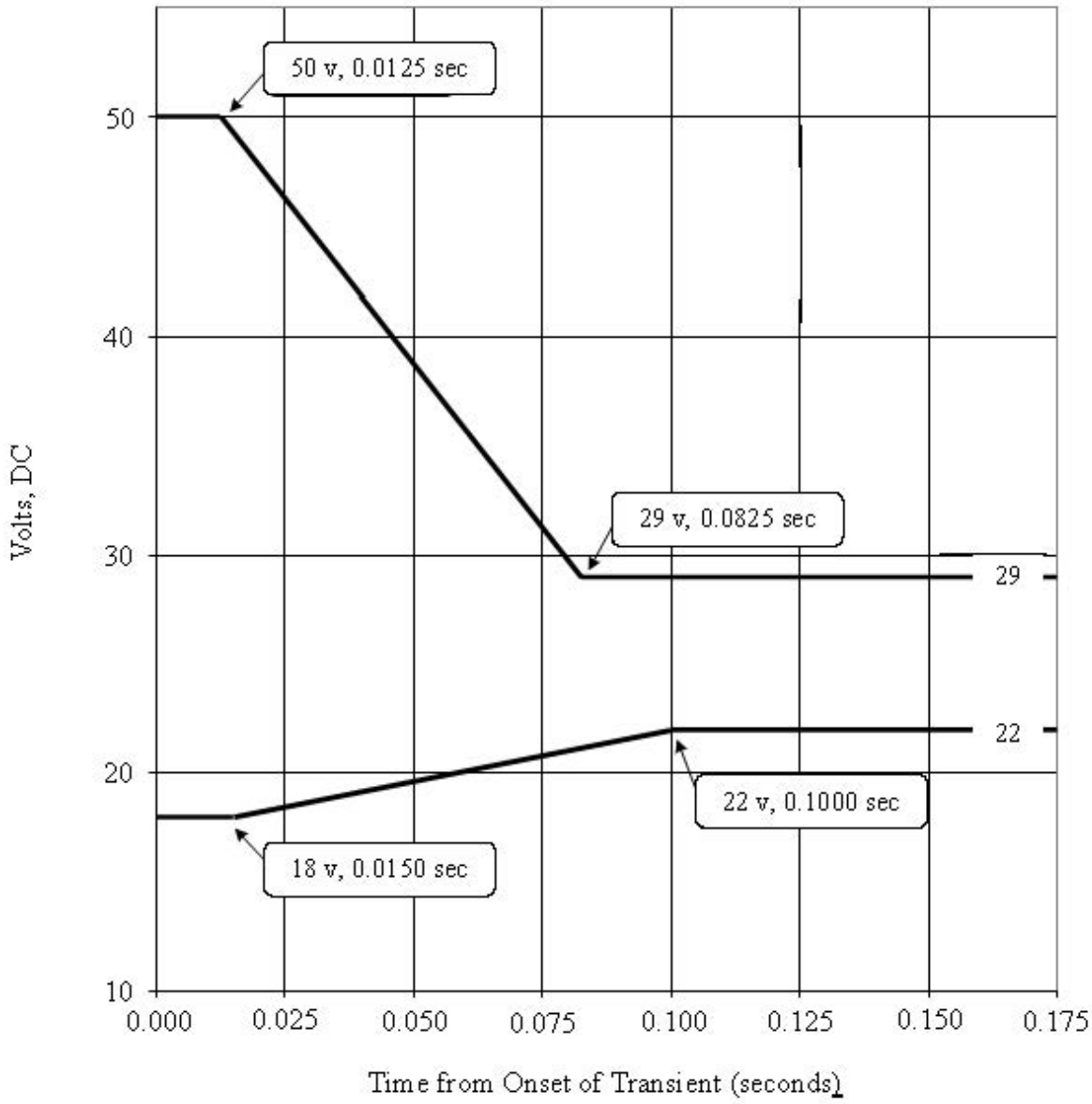


FIGURE 4. Input - Type DC converter envelope of normal voltage transients (see 3.5.1.2.3.1.1, 4.6.2.1).

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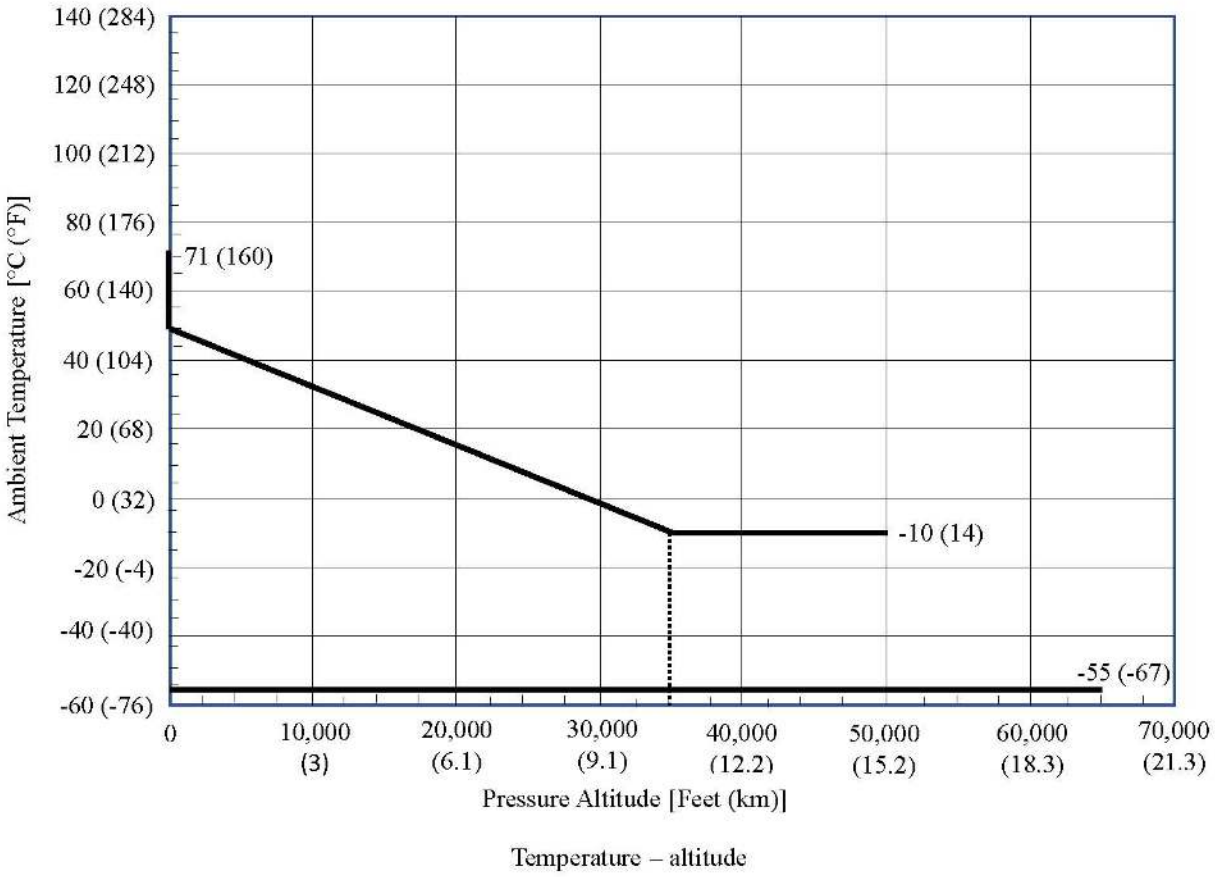


FIGURE 5. Temperature – altitude (see 3.5.2.1, 4.6.2.1).

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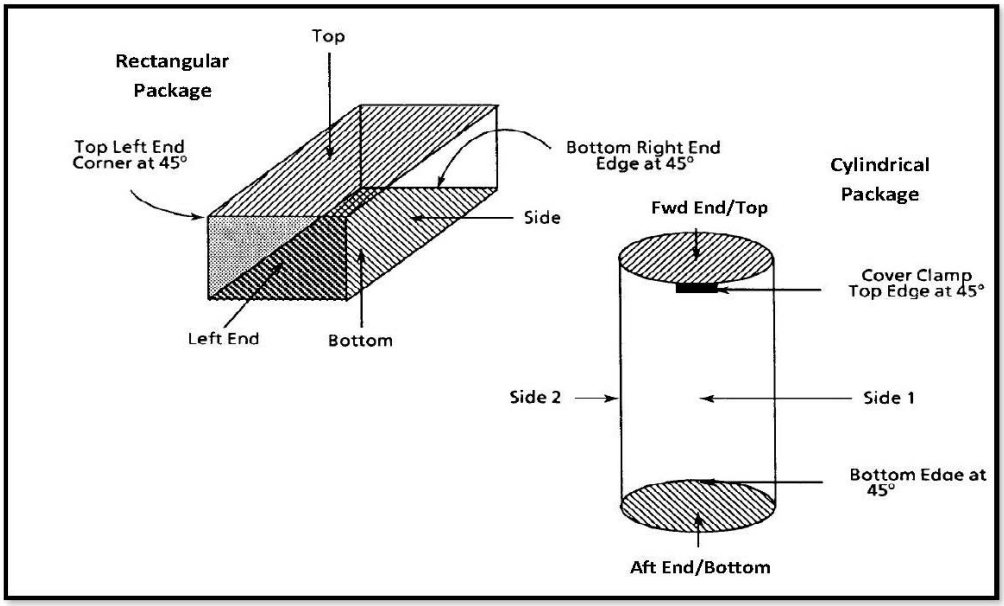


FIGURE 6. Standard drop orientations for rectangular and cylindrical packages (see 4.6.3.3).

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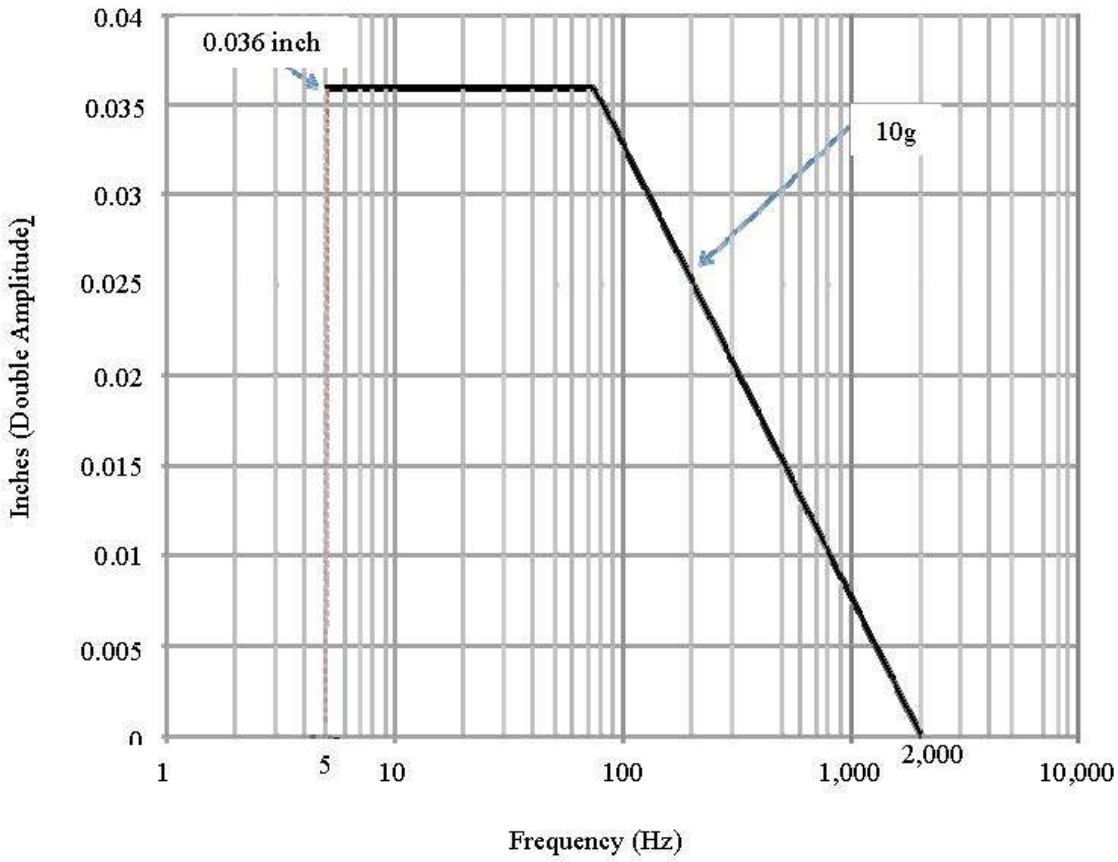


FIGURE 7. Sinusoidal vibration reference profile (see 4.6.3.4.2).

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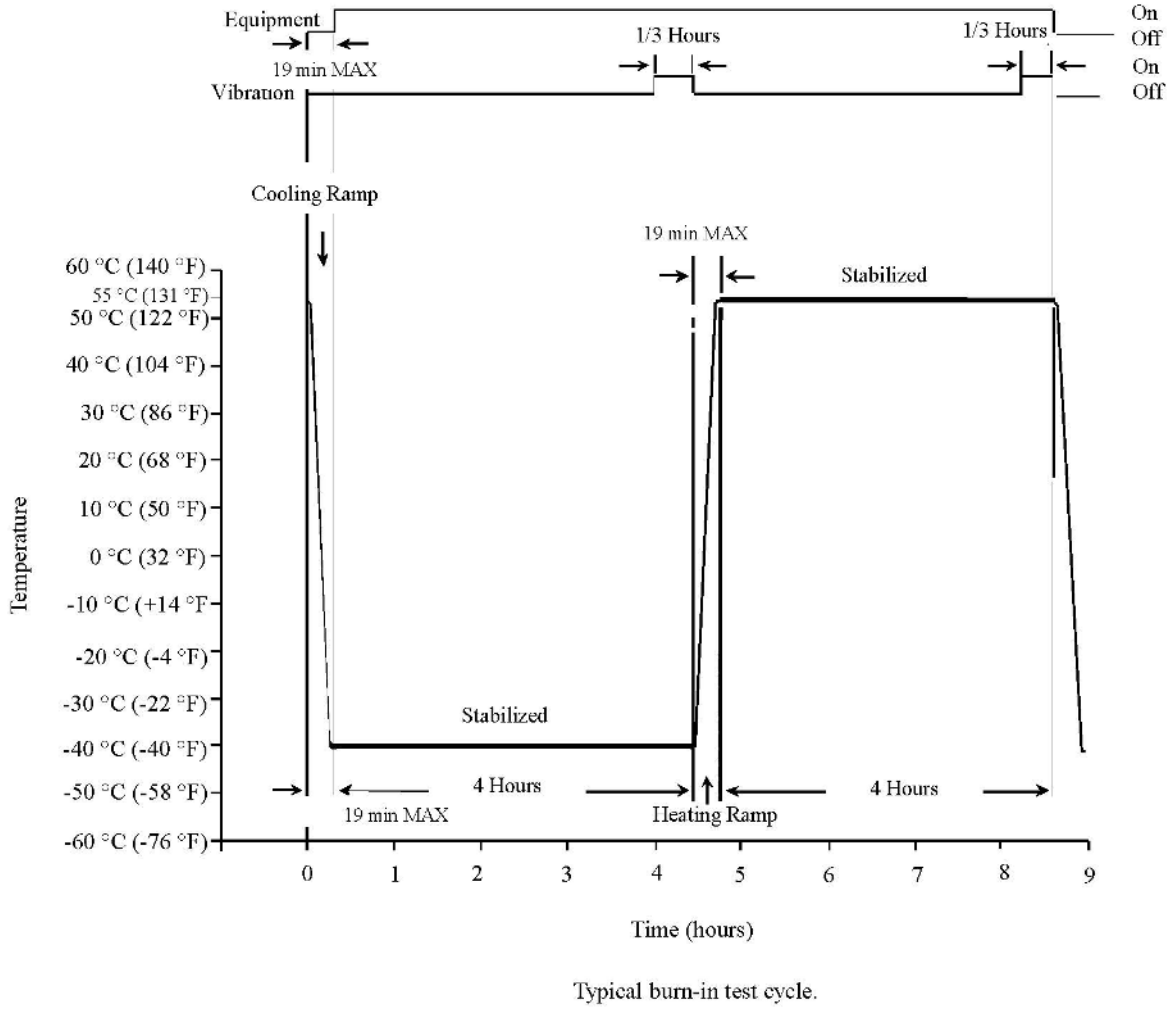


FIGURE 8. Typical burn-in test cycle (see 4.6.5).

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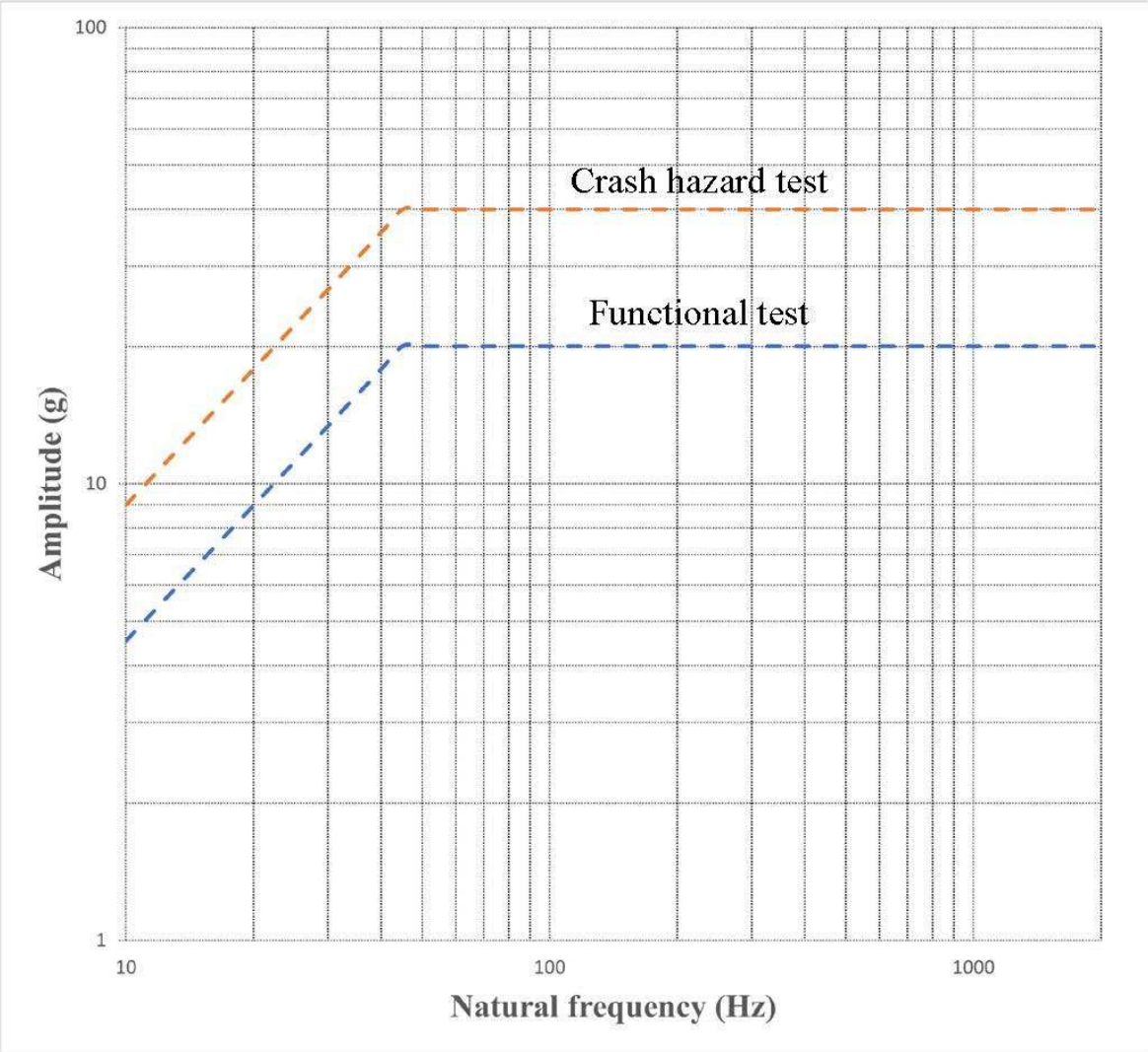


FIGURE 9. Functional and crash hazard shock test profile (see 4.6.3.3.1, 4.6.3.3.2).

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Custodians:
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Preparing activity:
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